

# BREAKING NEW GROUND IN ENERGY SUSTAINABILITY

The issue of climate change is posing an unprecedented challenge to the entire global community. Due to the buildup of greenhouse gases in the atmosphere, the climate of the Earth is changing quickly. Getting to net zero carbon emissions is a critical step in this direction.

The balance between greenhouse gas emissions and removal is known as net zero carbon emissions, which is frequently used interchangeably with the term carbon neutrality. In order to achieve balance, emissions must be reduced as much as feasible and any left-over emissions must be offset using a variety of technologies, such as carbon capture.

Achieving net zero carbon emissions is imperative because of the dangerous effects of climate change. We can prevent catastrophic effects by keeping global warming to 1.5 degrees Celsius over pre-industrial levels through the transition to a net-zero economy, as suggested by the Intergovernmental Panel on Climate Change (IPCC).

It is crucial to switch from fossil fuels to renewable energy sources including solar, wind, and hydropower. Governments and companies must make investments in sustainable energy infrastructure, encourage energy efficiency.

It is with this unwavering focus that the Bharat Petroleum Corporation Ltd (BPCL), a premier integrated energy company with a petroleum refining capacity of around 35.3 MMTPA, has pioneered technological advancement in its quest for net zero carbon emissions. The Maharatna, a Fortune Global company, has recently come up with two ground-breaking sustainable solutions to energy requirement – ethanol-diesel blend – for cleaner emissions, and high-efficiency LPG burner.

Research & Development is back bone for any company for giving leverage to the company for sustainable growth. Looking at the future energy landscape and Corporation's Net Zero 2040 mission, our corporate R&D programs are primarily focusing on developing innovative solutions for CO<sub>2</sub> capturing and utilization, Green H<sub>2</sub> and Biofuels and also leveraging R&D for supporting the existing business growth and sustainability. BPCL is aggressively investing in technological advancement in the processes and efficiently



BPCL, Kochi Refinery

monitor & optimize the operation of existing Assets through digital advancement, and collaborations with renowned institutes. This underscores the petrochemical marketing giant's sustainability agenda and Net Zero Mission, a pledge to create a better future. The emerging technologies in renewal and conventional areas both are vital to reduce fossil fuel uses on sustainable basis and readiness for the future energy.

The high-efficiency LPG burner developed by the Centre has revolutionized energy conservation. The use of this high-efficiency burner will save one LPG cylinder annually as it is said to be 10 per cent more efficient than the conventional burner. The additional advantage of the burner is that it is easy to retrofit in existing burners of similar dimensions.

The R&D Centre has successfully developed a PNG burner with an efficiency of 70%, surpassing the reported 55% efficiency achieved so far. The plan is to conduct a pilot and roll out the energy-efficient PNG stove in FY 2023-24, contributing toward reduction in import dependency and burden on the exchequer.

The creation of such ground-breaking technology must be accelerated by continued investment in research and

development (R&D). This includes developments in technologies for direct air capture (DAC), carbon capture and storage (CCS) and renewable energy. Governments, businesses, and academic institutions should work together to promote innovation and commercialize innovative technology.

Similarly, addressing the global challenge of climate change requires cooperation among countries. International agreements like the Paris Agreement provide a framework for collaboration, encouraging countries to set ambitious emissions reduction targets and support each other in achieving them. Sharing best practices, knowledge, and resources can expedite the transition to a net-zero future. No wonder, the Centre has risen to global prominence as a leading research centre with an enduring commitment to adopt an innovative approach to valorize bio-refinery waste.

Leading by example, more companies are aligning their operations and production processes for improving sustainability scores by leveraging innovation in technology that support simple shift or changeover procedures for maintaining an eco-friendly and thriving sustainable environment.